Safe Active Discrimination of Fractional, Delayed, and Finite Latent Memory in a Strong-Allee Predator–Prey Model

Ibrahim Alraddadi
Department of Mathematics, Faculty of Science
Islamic University of Madinah, Madinah, Saudi Arabia
ialraddadi@iu.edu.sa

August 17, 2026

Abstract

We study how controlled ecological experiments can distinguish Caputo memory from integer-order, delayed, and finite latent alternatives in a strong-Allee predator–prey system. The main text is organized around four analytical results and an expanded numerical study. Structurally, the collocated Caputo transfer has a noninteger high-frequency signature that cannot be exactly reproduced by the declared finite latent or retarded-delay classes. Practically, finite exponential mixtures approximate fractional memory over finite horizons, producing a complexity-dependent testing obstruction. Directly on the ecological prey channel, interval arithmetic certifies finite-state approximations whose error decreases from 0.5335 at $m = 4$ to 0.0070 at $m = 32$ for $\alpha = 0.85$, $A = 0.25$, and $T = 12$; under the declared pulse budget, the associated Gaussian testing lower bound approaches chance. We then introduce a combined fractional-delay model that reduces to the pure Caputo case at $\tau = 0$ and to the DDE case at $\alpha = 1$. New simulations show that delay systematically shifts the predator peak, erodes the Allee safety margin, and creates intermediate responses that are not well represented by either pure limit. Across six perturbation families and a large four-class benchmark, broadband inputs improve discrimination but cross the Allee threshold more frequently, whereas transient inputs are safer numerically but less informative. The results expose a fundamental safety–informativeness trade-off in experimental identification of ecological memory.

1 Introduction

Fractional derivatives are widely used to represent hereditary and nonlocal effects in dynamical systems [4, 16, 6, 11], including predator–prey models [1, 10, 18, 17]. The central empirical difficulty is that a fitted fractional order does not by itself identify a fractional mechanism: over finite horizons, delayed feedback and finite-dimensional latent states can reproduce similar trajectories. Experiment design for fractional-order identification already exists [13, 8, 20, 9], while safety-constrained information acquisition and safe active model discrimination are established general problems [19, 15]. We therefore focus on a narrower question:

> *Which perturbations and measurements can distinguish fractional memory from delayed or latent alternatives in a strong-Allee ecological system, and when does ecological safety make that discrimination fundamentally difficult?*

The ecological backbone is the validated strong-Allee predator–prey model used in our previous study [21, 5, 3, 12]. The present paper moves from certification of dynamics to *experimental discrimination of memory mechanisms*. The ODE and Caputo mechanisms share the same ecological equilibrium and Jacobian; retarded-delay and finite-state latent models are calibrated rival response laws compared through matched inputs, observations, and physical units.

Main contributions. The paper is organized around four contributions, rather than around a long sequence of standalone theorems.

1. **Structural and finite-horizon separation.** We show that the collocated Caputo transfer has a noninteger high-frequency signature that cannot be exactly reproduced by finite rational latent models or by the declared retarded-delay class. At the same time, finite exponential mixtures approximate the fractional kernel on a finite horizon, creating a complexity-dependent discrimination barrier.

2. **Certified prey-response testing obstruction.** Directly on the linearized ecological prey channel, a pole/branch decomposition yields interval-enclosed finite-state approximations. Under a declared pulse budget and Gaussian observations, the corresponding minimax testing lower bound approaches chance as latent complexity grows.

3. **Numerical validation and a new fractional-delay stress test.** In addition to the calibrated ODE/Caputo/DDE/latent benchmark, we introduce a combined fractional-delay model. It continuously connects the pure Caputo limit ($\tau = 0$) to the DDE limit ($\alpha = 1$), allowing direct simulation of how fractional order and delay jointly alter phase, transient amplitude, safety margin, and experimental discriminability.

4. **Safety–informativeness trade-off.** Across six input families, broadband excitation improves memory discrimination but increases the rate of crossing the strong-Allee threshold. Transient inputs are safer numerically but less informative, making the constrained experiment-design problem visible rather than merely theoretical.

The detailed proofs, secondary analytical results, Bayesian sequential extension, and solver derivations are moved to the Supplementary Material. The main paper retains only the analytical statements required to interpret the numerical results.

2 Ecological Model and Competing Memory Mechanisms

2.1 Strong-Allee predator–prey backbone

The nonlinear ecological model is

$$f_1(x, y) = rx\left(1 - \frac{x}{K}\right)\left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \tag{1}$$

$$f_2(x, y) = e\frac{axy}{1 + hx} - my, \tag{2}$$

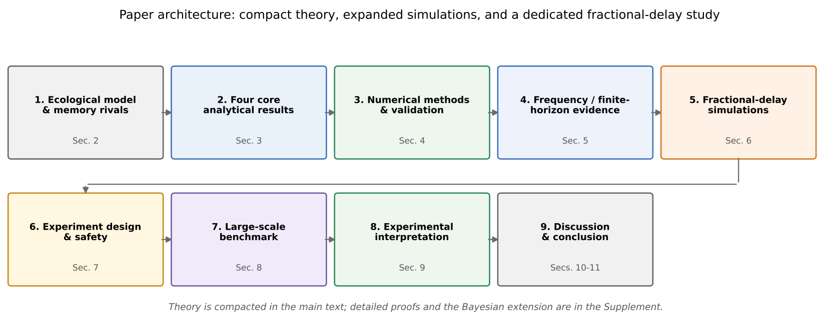

Figure 1: Paper structure after the Q1-focused restructuring: certified ecological model and rival memory laws, four essential analytical results, numerical validation, dedicated fractional-delay simulations, experiment design and safety analysis, and the large-scale discrimination benchmark. Detailed proofs and secondary theory are placed in the Supplementary Material.

with locked parameters

$$r = \frac{3}{2}, \qquad K = 1, \qquad a = 1, \qquad h = \frac{1}{2}, \qquad e = \frac{4}{5}, \qquad m = \frac{2}{5}.$$

The coexistence equilibrium and Jacobian are

$$x^* = \frac{2}{3}, \qquad y^*(A) = \frac{2(2-3A)}{9A},$$
$$J(A) = \begin{pmatrix} \dfrac{7A-2}{8A} & -\dfrac{1}{2} \\ \dfrac{2-3A}{10A} & 0 \end{pmatrix}.$$

For a commensurate Caputo model, local stability follows the Matignon sector condition [14]. In the complex-pair regime $2/7 < A < 10/19$, the critical order is

$$\alpha^*(A) = \frac{2}{\pi} \arctan\left(\frac{\sqrt{4D(A) - T(A)^2}}{T(A)} \right),$$

with $T(A) = \operatorname{tr} J$ and $D(A) = \det J$. At $A = 0.30$, $\alpha^* \approx 0.969012276$, so the fractional system with $\alpha = 0.9$ is stable although the integer-order limit is unstable [12]. For most discrimination experiments we use $A \leq 2/7$, where the common operating point remains in the stable regime.

2.2 Four rival mechanisms

Dimensional consistency is enforced with the reference time τ_0 [16, 6, 11]. Around the operating point, the four candidate mechanisms are

$$\begin{aligned} \text{ODE:} \quad & \dot{\xi} = J\xi + Bu, \\ \text{Caputo:} \quad & \tau_0^{\alpha-1} {}^C D_t^\alpha \xi = J\xi + Bu, \\ \text{DDE:} \quad & \dot{\xi} = A_0 \xi(t) + A_1 \xi(t-\tau) + Bu, \\ \text{latent:} \quad & \dot{q} = \bar{A} q + \bar{B} u, \qquad y = \bar{C} q. \end{aligned}$$

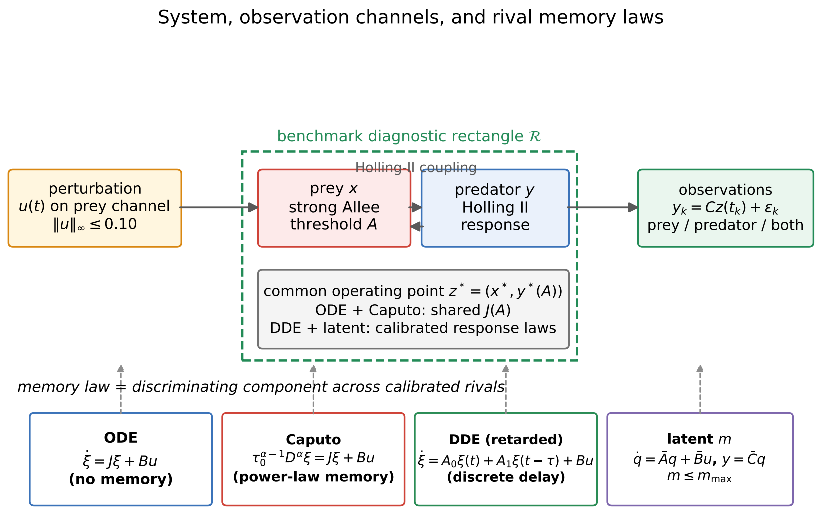

Figure 2: Ecological and experimental setting. The strong-Allee predator–prey system is perturbed on the prey channel and observed through prey, predator, or both populations. ODE and Caputo models share the exact ecological Jacobian; DDE and latent models are calibrated rival response laws. The dashed rectangle is a numerical benchmark diagnostic region, not an invariant-set certificate.

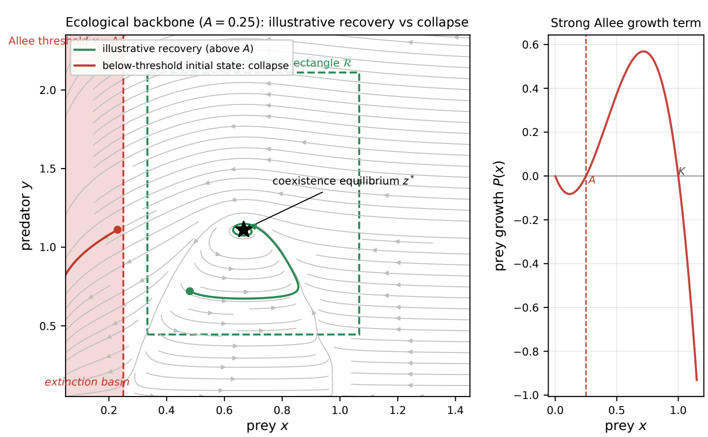

Figure 3: Strong-Allee backbone at $A = 0.25$. Left: representative recovery above the threshold and collapse from below A. Right: prey growth term $P(x)$ showing the sign change at the Allee threshold. This geometry explains why experiment design must account for state safety, not only actuator amplitude.

Only the Caputo mechanism carries a fractional order. The finite latent class is restricted by an explicit state-dimension budget m.

Combined fractional-delay stress-test model. To address the interaction of memory and explicit delay directly, the numerical study also uses

$$\tau_0^{\alpha-1\,C}D_t^\alpha x(t) = P(x(t);A) - \frac{ax(t)y(t)}{1+hx(t)} + u(t), \tag{3}$$

$$\tau_0^{\alpha-1\,C}D_t^\alpha y(t) = e\frac{ax(t-\tau)y(t)}{1+hx(t-\tau)} - my(t). \tag{4}$$

A constant equilibrium history is used on $[-\tau, 0]$. Equation (4) reduces to the pure Caputo model at $\tau = 0$ and to the retarded DDE benchmark at $\alpha = 1$. It is used as a numerical stress test, not as an additional headline theorem.

3 Analytical Backbone: Four Results Used by the Experiments

The complete derivations and secondary results are given in the Supplementary Material. Here we retain only the four statements that directly support the numerical study.

3.1 Result I: structural separation

For the collocated channel $CB \neq 0$, the Caputo transfer

$$G_\alpha(s) = C\big(q_\alpha(s)I - J\big)^{-1}B, \qquad q_\alpha(s) = \tau_0^{-1}(s\tau_0)^\alpha,$$

has the high-frequency expansion

$$G_\alpha(s) = (CB)\tau_0^{1-\alpha}s^{-\alpha} + O(|s|^{-2\alpha}).$$

Theorem 3.1 (Structural separation)**.** *For $0 < \alpha < 1$ and $CB \neq 0$, the Caputo transfer cannot equal any finite-dimensional rational latent transfer on a nonempty open subset of a common analytic domain. It also cannot equal a strictly proper finite-state retarded model with finitely many discrete delays, because the latter has $O(\omega^{-1})$ high-frequency decay whereas the Caputo channel has $O(\omega^{-\alpha})$ decay.*

The theorem explains why exact discrimination is possible in principle. It does not imply that finite noisy experiments are easy.

3.2 Result II: finite-horizon latent approximation

The Caputo memory kernel admits a diffusive representation and can be approximated on $[0, T]$ by a positive exponential sum. For a constructive m-mode approximation k_m,

$$\|k_\alpha - k_m\|_{L^1(0,T)} \leq \widehat{E}_m, \qquad \widehat{E}_m \downarrow 0$$

along an expanding nested rate hierarchy.

Theorem 3.2 (Finite-horizon approximation and complexity barrier)**.** *For every $T < \infty$ and $\varepsilon > 0$, there exists a finite positive exponential mixture whose $L^1(0,T)$ distance from the Caputo kernel is below ε. Consequently, without a finite complexity restriction on the latent rival, no fixed finite-horizon experiment can guarantee a uniform positive separation from the entire latent hierarchy.*

This is the analytical reason for treating latent dimension as an experimental-model complexity budget rather than an unconstrained nuisance parameter.

3.3 Result III: testing obstruction under a physical input budget

Let S be the observation operator, R the Gaussian covariance, and $C_{\mathrm{obs}} = \frac{1}{2}\|R^{-1/2}S\|^2$. Young's inequality converts the approximation bound into an observation-space bound.

Theorem 3.3 (Complexity-dependent testing obstruction)**.** *If the admissible experiment class satisfies $\|u\|_2 \leq B$, then the composite minimax error for testing the fractional response against a latent hierarchy containing an explicit competitor with error $\widehat{E}_m$ obeys*

$$P_e^*(m) \geq \Psi\Big(C_{\mathrm{obs}}\widehat{E}_m^2 B^2\Big),$$

where Ψ is any valid equal-covariance Gaussian two-point lower bound (Pinsker is used in the numerical plots). Thus $P_e^*(m) \to 1/2$ whenever $\widehat{E}_m \to 0$ and B remains bounded.

3.4 Result IV: ecological prey-response approximation

The ecological result works with the exact prey-to-prey fractional transfer, avoiding the invalid factorization $(s^\alpha I - J)^{-1} = (sI - J)^{-1}s^{-\alpha}$. A pole/branch decomposition realizes the principal-sheet pole pair exactly with a stable two-state integer-order block and approximates only the branch remainder.

Theorem 3.4 (Certified finite-state approximation of the prey response)**.** *For the displayed Matignon-stable cells and $m \in \{4, 8, 16, 32\}$, there exists an explicit stable finite-state prey-response surrogate g_m such that*

$$\|g_\alpha - g_m\|_{L^1(0,T)} \leq \widehat{E}_m^{\mathrm{state}},$$

where the bound is obtained by outward-rounded interval subdivision and a rigorous tail estimate. At $\alpha = 0.85$, $T = 12$, the certified enclosure at $A = 0.25$ decreases from 0.5335 ($m = 4$) to 0.0070 ($m = 32$).

For the declared pulse, $\|u\|_\infty \leq 0.10$ implies $\|u\|_2 \leq 0.120$. At $\sigma = 0.10$, the resulting Pinsker testing lower bounds at $A = 0.25$ are approximately $0.340, 0.462, 0.492, 0.498$ for $m = 4, 8, 16, 32$, respectively. Hence the finite-state rival becomes statistically indistinguishable from the fractional prey response as its allowed complexity grows.

4 Numerical Methods and Validation

The computational layer is designed to support the analytical claims rather than replace them. All scripts and machine-readable outputs are released under `benchmark/` and `figures/`.

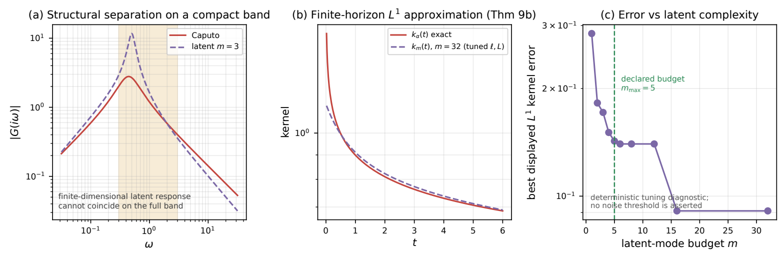

Figure 4: Exact structural separation and finite-horizon approximation are complementary, not contradictory. The Caputo transfer has a distinct noninteger signature, yet increasingly rich finite exponential mixtures track the fractional kernel over a finite horizon.

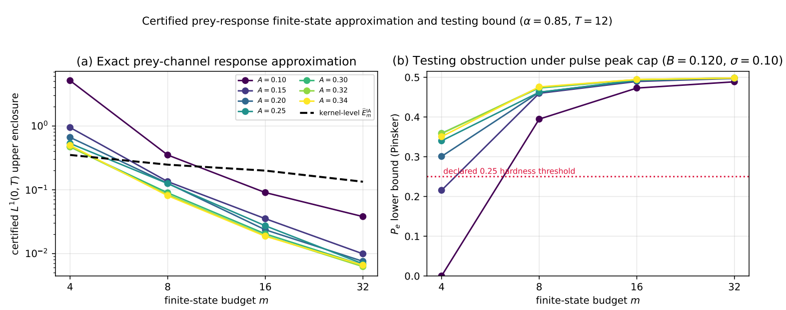

Figure 5: Certified ecological prey-response approximation and corresponding testing obstruction. Increasing the finite-state budget sharply reduces the interval-enclosed response error and drives the lower bound on testing error toward chance.

4.1 Solvers

ODE and latent-state systems are integrated with fourth-order Runge–Kutta. Retarded DDEs use an interpolated-history RK4 solver. Caputo systems use the fractional Adams–Bashforth–Moulton PECE scheme [16]; its scalar implementation is validated against the exact Mittag–Leffler solution.

For the combined fractional-delay model (4), we use the same PECE history convolution while evaluating the delayed prey term by interpolation of the already computed trajectory. For $\tau > 0$ all reported grids satisfy $\tau \geq h$, so the delayed value required by the corrector lies in known history. The two limiting cases provide strong numerical gates:

- $\tau = 0$ is delegated exactly to the validated Caputo solver;
- at $\alpha = 1$, the fractional-delay solver converges to the independent DDE RK4 solution.

At $A = 0.25$, $\tau = 0.35$, the maximum absolute discrepancy between the $\alpha = 1$ fractional-delay solver and the DDE solver is 6.75×10^{-6} at $N = 200$, 1.68×10^{-6} at $N = 400$, and 4.20×10^{-7} at $N = 800$. For the representative fractional-delay cell $(\alpha, \tau) = (0.85, 0.35)$, the max difference against the $N = 800$ solution drops from 2.18×10^{-3} at $N = 200$ to 8.05×10^{-4} at $N = 400$.

4.2 Inputs, observations, and metrics

Six bounded inputs are compared at the common peak-amplitude budget $\|u\|_\infty \leq 0.10$: pulse, multiscale pulse train, sinusoid, multisine, chirp, and pseudorandom binary sequence (PRBS). Observations use prey, predator, or both channels. Numerical safety is reported separately from solver non-divergence using the minimum Allee margin

$$\min_t (x(t) - A)$$

and the rate of threshold crossing.

For fractional-delay experiments we additionally report (i) trajectory distance to the pure Caputo limit, (ii) trajectory distance to the pure DDE limit, (iii) predator peak time and amplitude, and (iv) a robust bridge-discrimination score defined as the minimum of the distances to the two limiting explanations.

5 Frequency-Domain and Finite-Horizon Numerical Evidence

The structural results are visible before any large benchmark is run. Figure 6 shows transfer magnitudes on the prey channel at $A = 0.25$. Figure 7 shows the phase separation and the high-frequency convergence to $-\alpha\pi/2$ on collocated channels.

The finite-horizon approximation results then explain why frequency-domain separation does not automatically translate into easy model selection. The safe memory-discrimination atlas in Figure 8 displays the protocol-restricted lower bound on testing error across fractional order, Allee parameter, and latent complexity. Cells outside the focal Matignon-stable regime are explicitly excluded from the headline interpretation.

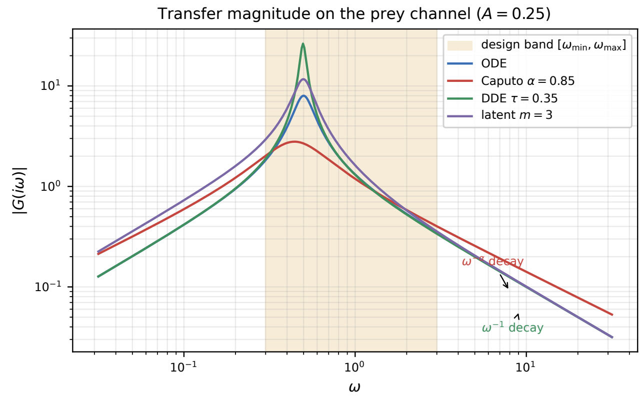

Figure 6: Transfer magnitude on the prey channel for the four memory mechanisms at $A = 0.25$. The fractional response separates across the design band and approaches its noninteger high-frequency decay.

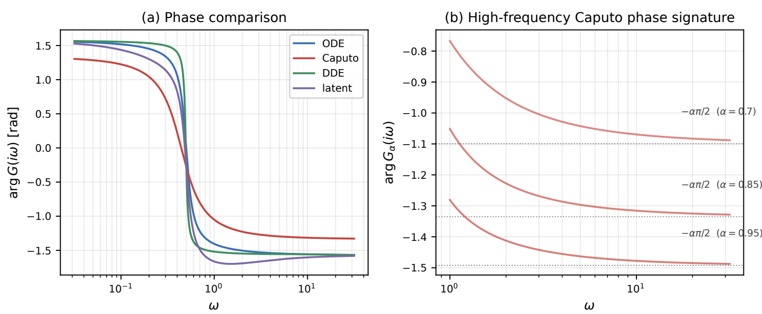

Figure 7: Phase evidence for the fractional signature. Left: Caputo, ODE, DDE, and finite latent responses across frequency. Right: the Caputo phase converges to the scale-free asymptote $-\alpha\pi/2$.

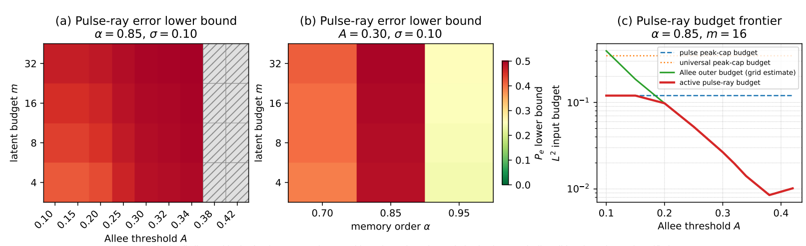

Figure 8: Protocol-restricted memory-discrimination atlas. Increasing latent complexity and moving toward weakly separated fractional regimes raise the lower bound on model-selection error. Hatched cells fall outside the focal Caputo stable regime.

6 Fractional-Delay Model: Dedicated Simulation Study

A natural robustness question motivates a dedicated numerical study of a mechanism carrying *both* fractional memory and explicit delay. Equation (4) is useful because its boundaries are exactly the two previously studied mechanisms: $\tau = 0$ gives the pure Caputo model, while $\alpha = 1$ gives the DDE benchmark. The interior of the (α, τ) plane therefore tests whether the two effects reinforce, compensate, or create ambiguous response signatures.

6.1 Transient trajectories

Figure 9 fixes $A = 0.25$, $\alpha = 0.85$, and applies the same prey pulse while increasing τ. The prey response changes moderately, whereas the predator response is visibly shifted and its late transient becomes increasingly delay-sensitive. Importantly, the trajectories remain above the Allee threshold for this pulse experiment.

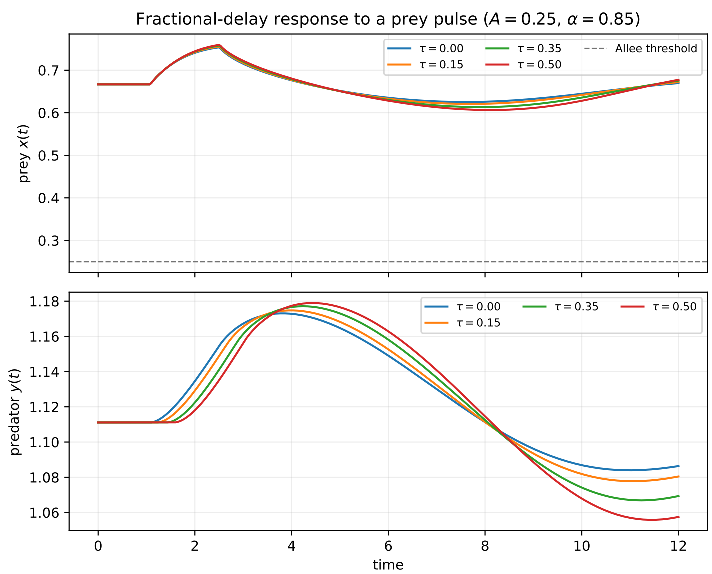

Figure 9: Combined fractional-delay response to the same prey pulse at $A = 0.25$ and $\alpha = 0.85$. Increasing delay shifts the predator response and changes the late transient while preserving the fractional-memory relaxation pattern.

6.2 Fractional-order/delay parameter plane

Figure 10 sweeps $\alpha \in [0.70, 1]$ and $\tau \in [0, 0.5]$. The left boundary has zero distance to the pure Caputo model by construction; the top boundary has zero distance to the pure DDE model. The interior quantifies how rapidly the combined mechanism departs from either limiting explanation. The safety panel shows that the same delay can be benign at lower α but erode the Allee margin as the system approaches the integer-order limit.

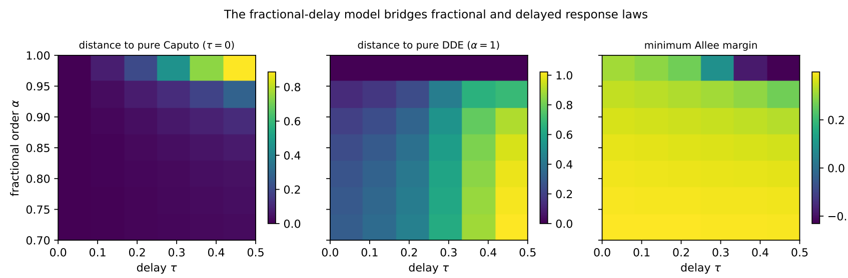

Figure 10: Fractional-delay bridge over (α, τ). Left: trajectory distance to the pure Caputo limit. Center: distance to the pure DDE limit. Right: minimum Allee margin under the same pulse. The interior of the parameter plane contains responses that are not well represented by either limit alone.

6.3 Which input exposes the combined mechanism?

For $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$, we score each input by the mean of the minimum trajectory distance to the pure Caputo and pure DDE explanations. Figure 11 shows that multisine excitation gives the largest bridge-discrimination score, but it crosses the Allee threshold in one third of the tested cells. PRBS, sinusoid, and chirp cross in every tested cell; pulse and multiscale excitation remain above threshold but are less discriminating. This reproduces the global safety–informativeness tension on a model that contains both kinds of memory simultaneously.

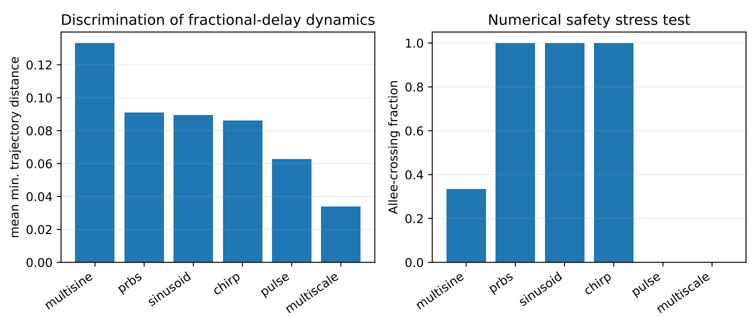

Figure 11: Input-design stress test for fractional-delay dynamics. Left: mean minimum trajectory distance from the two limiting explanations (pure Caputo and pure DDE). Right: numerical Allee-crossing fraction on the same (α, τ) grid. Broadband inputs reveal the combined mechanism more strongly but are substantially less safe.

6.4 Delay sweep

The delay sweep in Figure 12 makes the mechanism interpretable. Increasing τ shifts the predator peak later for every tested fractional order and increases its peak amplitude modestly. At the same time the minimum Allee margin decreases, especially near $\alpha = 1$. Thus the delay is visible not merely as an additional fitting parameter but as a reproducible shift in response timing and safety margin.

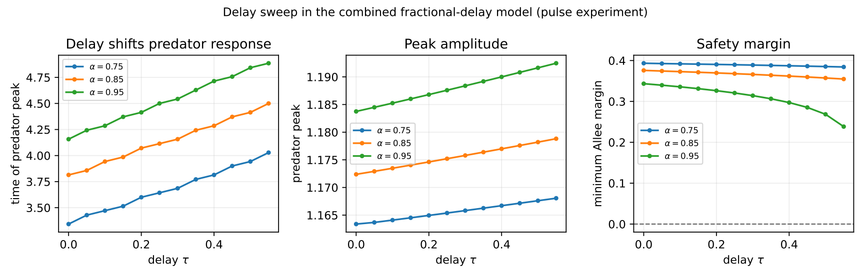

Figure 12: Delay sweep under the pulse experiment. The predator peak occurs later as τ grows, peak amplitude increases moderately, and the numerical Allee margin decreases. These trends are consistent across the three displayed fractional orders.

7 Experiment Design and Safety

The analytical design problem is summarized in the main paper and proved in full in the Supplementary Material. For two Gaussian linearized models with common covariance, the pairwise information criterion can be written

$$D_{ij}(u) = \frac{1}{2}\langle u, Q_{ij}u\rangle.$$

Under an energy constraint, the unconstrained optimum is the leading eigenfunction of Q_{ij}; for a finite robust frequency design, an optimum can be represented with finite frequency support. These standard design consequences are used here only to motivate the six implementable waveform families in Figure 13.

7.1 Strong-Allee safety

A one-sided Allee constraint requires

$$x(t) > A + \eta.$$

For the linearized system a sufficient pulse-amplitude envelope follows from the prey impulse gain; for the nonlinear system the strict inward-pointing rectangle theorem and its face inequalities are given in the Supplementary Material. Crucially, state safety alone does not impose a universal bound on input energy without a physical waveform/bandwidth class, because high-frequency inputs may be strongly attenuated by a stable strictly proper channel. This is why all quantitative testing bounds in the main paper are tied to an explicit actuator or protocol class.

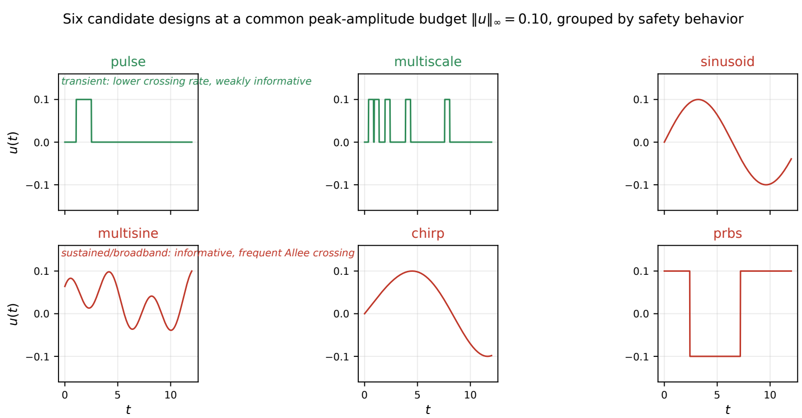

Figure 13: Six candidate perturbations at the common peak-amplitude budget. Transient and broadband designs probe the memory law differently and exhibit markedly different safety behavior in nonlinear simulations.

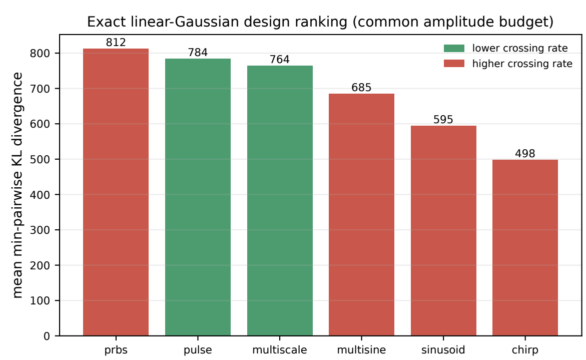

Figure 14: Linear-Gaussian design ranking by mean minimum pairwise KL divergence. This ranking ignores nonlinear state safety and therefore acts as an upper-level information benchmark rather than a final experimental recommendation.

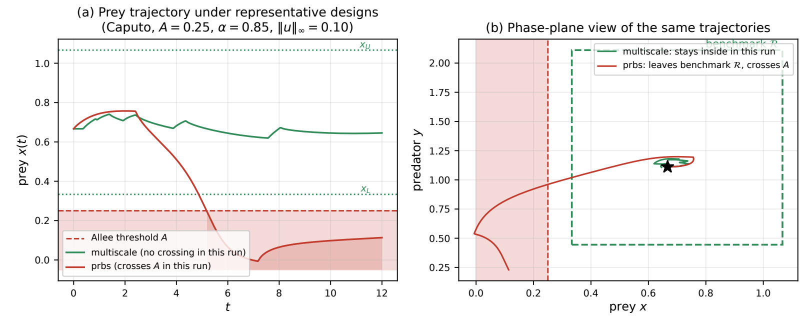

Figure 15: Representative nonlinear safety geometry. A multiscale transient remains above the Allee threshold in the shown run, while PRBS crosses it. The benchmark rectangle is a diagnostic region; rigorous positive invariance requires the strict face conditions stated in the Supplementary Material.

8 Large-Scale Model-Discrimination Benchmark

The calibrated benchmark compares ODE, Caputo, retarded DDE, and finite latent mechanisms at matched operating conditions. The primary stable-regime task uses $A \in \{0.20, 0.25\}$; a separate $A > 2/7$ task probes the easier stability-verdict problem. The decision rule is BIC, not Bayesian model evidence.

The full-factorial benchmark contains 200 replicates per cell and no solver divergences. In the primary four-class stable task, macro-averaged accuracy is 0.537 (chance 0.25). The design ranking is

$$\text{PRBS } (0.584) \approx \text{ multisine } (0.583) > \text{sinusoid } (0.534) > \text{chirp } (0.507) > \text{pulse } (0.414) > \text{multiscale } (0.324).$$

Discrimination degrades sharply as $\alpha \to 1$: accuracy falls from approximately 0.70 at $\alpha = 0.70$ to 0.31 at $\alpha = 0.95$. It is also noise-limited: 0.76 at high SNR, 0.44 at medium SNR, and 0.27 at low SNR. Observing both species performs best, although prey-only observation remains structurally sufficient.

8.1 Safety–informativeness trade-off

Numerical state safety is reported separately from information. At the common peak-amplitude budget, multiscale excitation has zero Allee crossings in the stable benchmark but the weakest discrimination. PRBS and sinusoidal excitation have the strongest discrimination among the tested families but cross the Allee threshold in essentially all stable-regime cells. Multisine occupies an intermediate position: high discrimination with a lower, but still non-negligible, crossing rate.

9 Prospective Experimental Interpretation

The numerical results translate into a practical microcosm protocol. A useful first experiment should (i) operate in a stable strong-Allee regime, (ii) apply a bounded prey perturbation, (iii) observe both populations when feasible, and (iv) combine a safe transient excitation with a second, more broadband perturbation only if

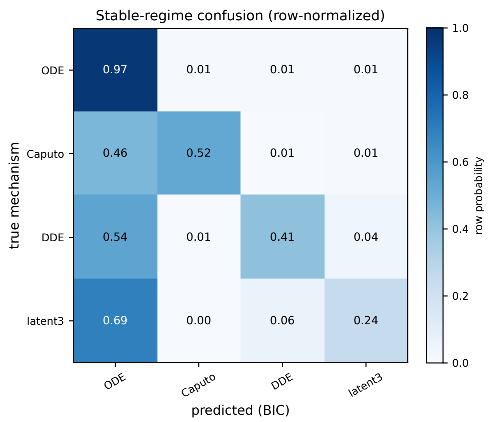

Figure 16: Row-normalized confusion matrix for the stable four-class task. The simplest ODE mechanism is frequently selected under weak excitation, whereas Caputo predictions are highly precise when selected.

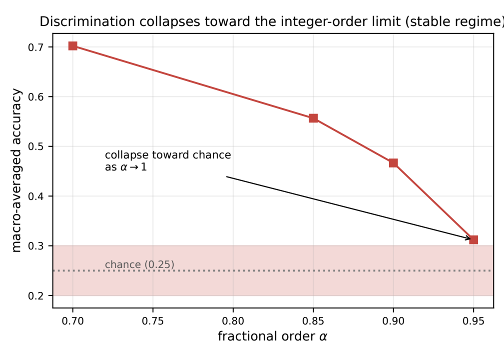

Figure 17: Model-selection accuracy versus fractional order. As α approaches the integer-order limit, the memory mechanisms become increasingly difficult to distinguish.

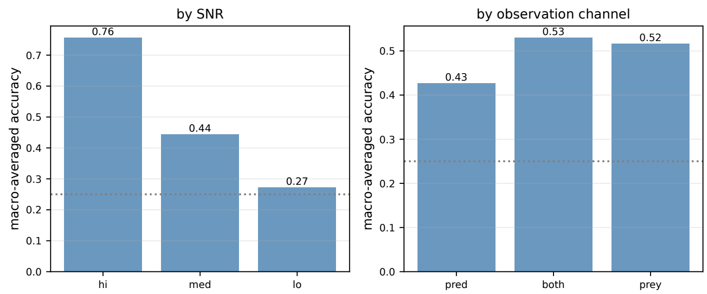

Figure 18: Accuracy stratified by SNR and observation channel. The benchmark is strongly noise-limited; both-species observation provides the highest average accuracy.

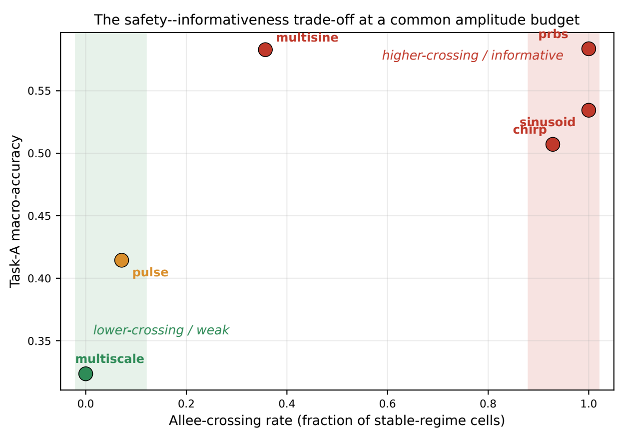

Figure 19: Headline benchmark trade-off. Broadband/sustained designs move upward in discrimination but also toward larger Allee-crossing rates; transient designs occupy the lower-crossing but less-informative region.

the first experiment leaves sufficient safety margin. Laboratory predator–prey microcosms provide a natural setting for such controlled perturbations [7, 22, 2].

The fractional-delay study adds a concrete diagnostic: delayed feedback should shift the timing of the predator peak, whereas decreasing fractional order primarily modifies the long transient and phase response. A design that records dense observations around the predator peak and continues sufficiently long to resolve the fractional tail is therefore more informative than uniform sparse sampling alone.

A Bayesian sequential layer is specified in the Supplementary Material but is not executed here. The present paper deliberately keeps the main evidence reproducible with deterministic solvers, interval enclosures, and BIC-based simulation benchmarks.

10 Discussion

The restructured results support three conclusions.

First, *structural identifiability and practical discrimination are different questions.* The Caputo transfer has an exact noninteger signature, yet finite latent models approximate the same response increasingly well on finite horizons. The testing bounds quantify this gap and explain why a flexible latent alternative can remain difficult even when the underlying model classes are analytically distinct.

Second, *delay and fractional memory should not be treated as mutually exclusive numerical explanations.* The combined fractional-delay simulations show a continuous bridge between the pure Caputo and pure DDE limits. Delay systematically shifts the predator response, while the fractional order controls the broader relaxation structure. The interior of the (α, τ) plane contains dynamics that can be poorly represented by either pure limit, which is precisely why direct numerical stress testing is valuable.

Third, *the most informative perturbation is not automatically the safest.* The same conclusion appears in the original four-class benchmark and in the new fractional-delay experiments. Broadband inputs enlarge response differences but more often drive prey below the strong-Allee threshold. This tension is ecological, not merely numerical: crossing A changes the long-term basin structure and invalidates an experiment designed only around local identifiability.

Limitations. The certified ecological response theorem is linearized and prey-channel specific; a full-state hierarchy-wide safety closure remains open. The fractional-delay study is numerical and uses a constant equilibrium history, one delayed trophic pathway, and a targeted parameter grid. The benchmark uses BIC rather than fully Bayesian evidence and does not execute adaptive design. These restrictions are stated explicitly to separate proved results from computational evidence.

Relation to prior work. General safe information limits and safe active model discrimination are already established [19, 15]; the contribution here is the interaction of finite-horizon fractional approximation, bounded latent complexity, strong-Allee ecological geometry, and direct numerical stress tests of the combined fractional-delay mechanism. The paper therefore does not claim novelty for safe experimental design in the abstract.

11 Conclusion

We developed a compact theory-and-simulation framework for discriminating ecological memory mechanisms under strong-Allee safety constraints. Four analytical results are retained in the main text: structural separation, finite-horizon latent approximation, a complexity-dependent testing obstruction, and an interval-certified finite-state approximation of the ecological prey response. The numerical evidence is substantially broader: frequency-response diagnostics, a dedicated combined fractional-delay simulation study, six perturbation families, safety metrics, and a large four-class discrimination benchmark.

The main empirical message is consistent across these layers. Fractional memory is structurally distinct, but finite experiments may not separate it from delayed or sufficiently rich latent alternatives. Increasing excitation improves discrimination, yet the strongest inputs are also the most likely to cross the Allee threshold. The appropriate next experimental question is therefore not simply *which model fits best*, but *which safe perturbation makes the competing memory mechanisms observably different*.

References

[1] Elsayed Ahmed, A. M. A. El-Sayed, and Hala A. A. El-Saka. Equilibrium points, stability and numerical solutions of fractional-order predator–prey and rabies models. *Journal of Mathematical Analysis and Applications*, 325(1):542–553, 2007.

[2] Lutz Becks, Frank M. Hilker, Horst Malchow, Klaus Jürgens, and Hartmut Arndt. Experimental demonstration of chaos in a microbial food web. *Nature*, 435:1226–1229, 2005.

[3] Luděk Berec, Elena Angulo, and Franck Courchamp. Multiple allee effects and population management. *Trends in Ecology & Evolution*, 22(4):185–191, 2007.

[4] Michele Caputo. Linear models of dissipation whose Q is almost frequency independent—ii. *Geophysical Journal International*, 13(5):529–539, 1967.

[5] Franck Courchamp, Tim Clutton-Brock, and Bryan Grenfell. Inverse density dependence and the allee effect. *Trends in Ecology & Evolution*, 14(10):405–410, 1999.

[6] Kai Diethelm. *The Analysis of Fractional Differential Equations: An Application-Oriented Exposition Using Differential Operators of Caputo Type*, volume 2004 of *Lecture Notes in Mathematics*. Springer, 2010.

[7] Gregor F. Fussmann, Stephen P. Ellner, Kyle W. Shertzer, and Nelson G. Jr. Hairston. Crossing the hopf bifurcation in a live predator–prey system. *Science*, 290(5495):1358–1360, 2000.

[8] Wiktor Jakowluk. Optimal input signal design for fractional-order system identification. *Bulletin of the Polish Academy of Sciences: Technical Sciences*, 67(1):37–44, 2019.

[9] Wiktor Jakowluk and Mirosław Świercz. Optimal input design for fractional-order system identification using an LMI-based frequency error criterion. *Applied Sciences*, 15(23):12665, 2025.

[10] Mohammad Javidi and Nasser Nyamoradi. Dynamic analysis of a fractional order prey–predator interaction with harvesting. *Applied Mathematical Modelling*, 37(20–21):8946–8956, 2013.

[11] Anatoly A. Kilbas, Hari M. Srivastava, and Juan J. Trujillo. *Theory and Applications of Fractional Differential Equations*, volume 204 of *North-Holland Mathematics Studies*. Elsevier, 2006.

[12] Maximiliano Lucius. Computer-assisted stability and extinction certificates for a caputo predator–prey system with a strong prey allee effect. Companion validated manuscript, draft dated 2026-07-30; published as Zenodo preprint DOI 10.5281/zenodo.21809908 (companion project fractional-memory-identifiability), 2026.

[13] Rachid Malti, Mohamed Mayoufi, and Stéphane Victor. Experiment design for elementary fractional models. *Communications in Nonlinear Science and Numerical Simulation*, 110:106337, 2022.

[14] Denis Matignon. Stability results for fractional differential equations with applications to control processing. In *Computational Engineering in Systems Applications*, volume 2, pages 963–968, Lille, France, 1996.

[15] Xinpei Ni, Melkior Ornik, Glen Chou, and Samuel Coogan. Safe, real-time active model discrimination and fault diagnosis for nonlinear systems via differentiable reachability, 2026. arXiv:2606.19590.

[16] Igor Podlubny. *Fractional Differential Equations*, volume 198 of *Mathematics in Science and Engineering*. Academic Press, 1999.

[17] G. Ranjith Kumar and K. Ramesh. Dynamical analysis of fractional-order bazykin's model with prey refuge, gestation delay and density-dependent mortality rate. *Iranian Journal of Science*, 49(1):79–91, 2024.

[18] Fathalla A. Rihan. Fractional-order delay differential equations with predator–prey systems. In *Delay Differential Equations and Applications to Biology*. Springer Singapore, 2021.

[19] Venkatesh Saligrama. When can safe controllers adapt? information before commitment, 2026. arXiv:2607.16895.

[20] Àngela Sebastià Bargues, José-Luis Polo Sanz, Irene García-Camacha Gutiérrez, and Raúl Martín Martín. Practical implementation of optimal experimental design using the fractional-order fricke–morse bioimpedance model. *Chaos, Solitons & Fractals*, 170:113374, 2023.

[21] Philip A. Stephens, William J. Sutherland, and Robert P. Freckleton. What is the allee effect? *Oikos*, 87(1):185–190, 1999.

[22] Takehito Yoshida, Lee E. Jones, Stephen P. Ellner, Gregor F. Fussmann, and Nelson G. Jr. Hairston. Rapid evolution drives ecological dynamics in a predator–prey system. *Nature*, 424:303–306, 2003.